Aarav Mehta

AI / Machine Learning Engineer (LLM & RAG Systems) | Tailored for VertexScale AI Labs (JD #1 — 96% Verified Match)

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EXECUTIVE SUMMARY — GROUNDED IN PORTFOLIO RECORD [SUM-02]

AI/ML Engineer specializing in Retrieval-Augmented Generation (RAG), Google Gemini & Transformer fine-tuning, and low-latency vector search pipelines. Experienced in deploying deterministic, citation-grounded LLM workflows with Python, PyTorch, LangChain, ChromaDB, and FastAPI, achieving 91.4% retrieval precision and 63% latency reduction on enterprise knowledge bases.

VERIFIED TECHNICAL COMPETENCIES — MATCHED TO TARGET JD (96% JD MATCH)

Competency Category	Selected Portfolio Skills & Proficiency Levels
Programming Languages	Python (Expert), SQL (PostgreSQL, SQLite) (Advanced)
AI & Machine Learning	Google Gemini API & Vertex AI (Structured Outputs) (Expert), PyTorch & Hugging Face Transformers (Advanced), RAG Pipelines (LangChain, LlamaIndex, FAISS, ChromaDB) (Expert), Scikit-Learn, Pandas, NumPy & OpenCV (Advanced)
Web & Backend Frameworks	FastAPI & Flask (Async Microservices) (Expert)
Cloud & DevOps	Docker, Kubernetes & GitHub Actions CI/CD (Advanced), AWS (ECS, EC2, S3, RDS) & Google Cloud Run (Advanced), Redis Caching & Celery Distributed Queues (Advanced)
Soft Skills	Cross-Functional Technical Communication (Expert)

PROFESSIONAL & INTERNSHIP EXPERIENCE

Software & AI Systems Engineer — NexaCloud Technologies Pvt. Ltd. (Full-Time) | Jul 2024 – Present | Mumbai, India [ID: EXP-01]

- Engineered production LLM microservice pipelines using Google Gemini API, Python, and FastAPI to automate customer ticket triage and contract extraction for 85+ enterprise clients.
- Reduced average ticket resolution time by 48% and saved 1,200+ manual review hours per month through schema-enforced AI extraction workflows.
- Built interactive React + TypeScript analytics dashboards and PostgreSQL audit logs with automated Google Sheets reporting exports.
- Mentored 4 junior developers on async Python patterns, unit testing (Pytest), and Docker containerization.

Machine Learning Engineering Intern — Quantiphi Analytics (Internship) | Jan 2024 – Jun 2024 | Mumbai, India [ID: EXP-02]

- Developed domain-specific Named Entity Recognition (NER) and document classification models using PyTorch and Hugging Face Transformers with 93.2% F1-score.
- Built automated data ingestion and evaluation pipelines processing 250GB+ of unstructured financial PDFs and earnings transcripts.
- Collaborated with MLOps engineers to containerize inference services on Google Cloud Run, cutting cloud compute costs by 27%.

SELECTED PRODUCTION PROJECTS

DocuPulse AI — Multi-Modal Enterprise RAG & Citation Engine — Lead AI Architect | Aug 2025 – Feb 2026 | Stack: Python, Gemini 2.5 API, FastAPI, ChromaDB, PyTorch, React, Docker [ID: PRJ-01]

- Architected a zero-hallucination Retrieval-Augmented Generation (RAG) engine indexing 50,000+ technical PDFs with hybrid BM25 + dense vector embeddings.
- Implemented strict primary-key citation verification using Gemini Structured Outputs, reducing unsupported claims from 14.0% to under 0.8%.
- Optimized async FastAPI inference endpoints with Redis semantic caching, cutting p95 query latency from 3.8s to 1.4s (63% reduction).

MedVision Triage — Chest X-Ray Anomaly Detection & Clinical Report Generator — ML Researcher | Jan 2024 – Jun 2024 | Stack: PyTorch, Hugging Face, Vision-Language Models, FastAPI, OpenCV, SQLite [ID: PRJ-03]

- Trained an ensemble DenseNet-121 and Vision-Language pipeline on 112,000+ NIH ChestX-ray images achieving 0.894 macro AUROC across 14 thoracic pathologies.
- Built a clinician-in-the-loop FastAPI web interface that generates structured radiology draft reports with Grad-CAM visual explainability heatmaps.

EDUCATION, CERTIFICATIONS, PUBLICATIONS & HONORS

Bachelor of Technology (B.Tech) in Computer Engineering (Honors Minor in Artificial Intelligence) — Veermata Jijabai Technological Institute (VJTI), Mumbai (Aug 2020 – May 2024) | **Grade: 9.32 / 10.0 CGPA** [ID: EDU-01]

Dean's Merit List (All 8 Semesters); Lead of AI & Robotics Society; Coursework: Deep Learning, NLP, Distributed Systems, DBMS, Algorithms

- **Google Cloud Certified — Professional Machine Learning Engineer** — Google Cloud (Apr 2025, Credential ID: GCP-PMLE-884120) [ID: CRT-01]
- **Generative AI with Large Language Models Specialization** — DeepLearning.AI & AWS (Jun 2024, Credential ID: DLAI-LLM-339104) [ID: CRT-03]
- **Research Paper:** "Citation-Grounded Retrieval-Augmented Generation for Low-Resource Indian Legal & Financial Documents" — IEEE International Conference on Data Science and Advanced Analytics (ICDSAA) (Apr 2024). Proposed a hybrid lexical-semantic chunking and primary-key verification method that reduces LLM hallucination rates by 38% on domain-specific PDFs. [ID: PUB-01]
- **Technical Speaker & Open-Source Maintainer:** "Building Deterministic AI Workflows with Gemini Structured Outputs and FastAPI" — Google Developer Group (GDG) Mumbai DevFest (Nov 2025). Delivered a hands-on workshop to 350+ developers on schema-guided generation, SQLite/Sheets synchronization, and zero-hallucination guardrails. [ID: PUB-02]
- **National Winner (1st Place) — Smart India Hackathon (AI & FinTech Track)** — Ministry of Education & AICTE, Government of India (Dec 2023): Led a 6-member team to build an AI-powered multilingual grievance redressal and document verification platform in 36 hours out of 1,400+ competing teams. [ID: ACH-01]
- **Top 15 Global Finalist — Google GenAI Exchange Hackathon** — Google Developer Groups & Startup India (Sep 2025): Awarded Best Grounded Enterprise RAG Implementation for DocuPulse AI among 4,500+ participants nationwide. [ID: ACH-02]
- **Institute Academic Excellence Award (Top 2% of Graduating Batch)** — VJTI Mumbai (May 2024): Recognized for graduating with a 9.32 CGPA and publishing peer-reviewed AI research during undergraduate studies. [ID: ACH-03]

Languages & Core Competencies: English (Native / Bilingual (TOEFL 114/120)), Hindi (Native / Bilingual), AI Safety & Grounded RAG Evaluation (Expert (Zero-Hallucination Prompting, Schema Validation))